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| 9 | (a) light-emitting diode (<i>allow LED</i>) | B1 | [1] |
| | (b) gives a high or a low output / +5 V or –5 V output
dependent on which of the inputs is at a higher potential | M1
A1 | [2] |
| | (c) (i) provides a reference/constant potential | B1 | [1] |
| | (ii) determines temperature of 'switch-over' | B1 | [1] |
| | (d) (i) relay | A1 | [1] |
| | (ii) relay connected correctly for op-amp output and high-voltage circuit
diode with correct polarity in output from op-amp | B1
B1 | [2] |
| 10 | (a) background reading = 19 | B1 | [1] |